

## Multiple Choice (10 marks)

1. The sign of the charge carriers in a doped semiconductor can be deduced by measuring which of the following properties?
  - (a) Magnetic susceptibility
  - (b) Hall coefficient
  - (c) Electrical resistivity
  - (d) Thermal conductivity
2. A single-electron atom has the electron in the  $l = 2$  state. The number of allowed values of the quantum number  $m_l$  is
  - (a) 5
  - (b) 4
  - (c) 3
  - (d) 2
3. A rod measures 1.00 m in its rest system. How fast must an observer move parallel to the rod to measure its length to be 0.80 m?
  - (a) 0.50 c
  - (b) 0.60 c
  - (c) 0.70 c
  - (d) 0.80 c
4. De Broglie hypothesized that the linear momentum and wavelength of a free massive particle are related by which of the following constants?
  - (a) Planck's constant
  - (b) Boltzmann's constant
  - (c) The Rydberg constant
  - (d) The speed of light
5. A refracting telescope consists of two converging lenses separated by 100 cm. The eyepiece lens has a focal length of 20 cm. The angular magnification of the telescope is
  - (a) 4
  - (b) 5
  - (c) 6
  - (d) 20

## True / False (10 marks)

*Answer True or False*

6. True or False: Fermions have antisymmetric wave functions and obey the Pauli exclusion principle, preventing them from occupying the same quantum state.
7. True or False: A particle moving at  $0.60c$  will travel only 288 m before decaying, based on its rest frame decay time of 2.0 ms.
8. True or False: The angular magnification of the refracting telescope with a 20 cm focal length eye-piece is 4.
9. True or False: The electrons in metals form a degenerate Fermi gas, leading to a higher mean kinetic energy than  $kT$ .
10. True or False: If the absolute temperature of a blackbody is increased by a factor of 3, the energy radiated per second per unit area decreases by a factor of 9.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss how to calculate the total spin quantum number of the electrons in the ground state of neutral nitrogen ( $Z = 7$ ) and explain its significance in quantum mechanics.
12. Discuss the electron configuration of an atom with filled  $n = 1$  and  $n = 2$  energy levels. How many electrons does it possess, and what implications does this have for its chemical properties?

*End of Paper*